

Đáp Án Bài Tập Cuối Chương

src/redux/slices/bookSlice.js

```
import { createSlice } from '@reduxjs/toolkit';

const bookSlice = createSlice({
  name: 'books',
  initialState: {
    books: [],
    filter: 'all',
  },
  reducers: {
    addBook: (state, action) => {
      state.books.push(action.payload);
    },
    removeBook: (state, action) => {
      state.books = state.books.filter(book => book.id !== action.payload);
    },
    toggleBorrowed: (state, action) => {
      const { bookId, memberId } = action.payload;
      const book = state.books.find(book => book.id === bookId);
      if (book) {
        book.borrowed = memberId !== null;
        book.borrowedBy = memberId;
      }
    },
    setFilter: (state, action) => {
      state.filter = action.payload;
    },
  },
});

export const { addBook, removeBook, toggleBorrowed, setFilter } = bookSlice.actions;
export default bookSlice.reducer;
```

src/redux/slices/memberSlice.js

```
import { createSlice } from '@reduxjs/toolkit';

const memberSlice = createSlice({
  name: 'members',
  initialState: {
    members: [],
  },
  reducers: {
    addMember: (state, action) => {
      state.members.push(action.payload);
    },
    removeMember: (state, action) => {
      state.members = state.members.filter(member => member.id !== action.payload);
    },
  },
});

export const { addMember, removeMember } = memberSlice.actions;
export default memberSlice.reducer;
```

src/redux/store.js

```
import { configureStore } from '@reduxjs/toolkit';
import bookReducer from './slices/bookSlice';
import memberReducer from './slices/memberSlice';

const store = configureStore({
  reducer: {
    books: bookReducer,
    members: memberReducer,
  },
});

export default store;
```

src/main.jsx

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import { Provider } from 'react-redux';
import store from './redux/store';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <Provider store={store}>
      <App />
    </Provider>
  </React.StrictMode>
);
```

src/App.jsx

```
import { BrowserRouter, Routes, Route } from 'react-router-dom';
import NavBar from './components/NavBar/NavBar';
import Home from './screens/Home/Home';
import BookDetail from './screens/BookDetail/BookDetail';
import MemberDetail from './screens/MemberDetail/MemberDetail';
import NotFound from './screens/NotFound/NotFound';

function App() {
  return (
    <BrowserRouter>
      <NavBar />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/book/:id" element={<BookDetail />} />
        <Route path="/member/:id" element={<MemberDetail />} />
        <Route path="*" element={<NotFound />} />
      </Routes>
    </BrowserRouter>
  );
}

export default App;
```

src/components/NavBar/NavBar.jsx

```
import { Link } from 'react-router-dom';
import './NavBar.css';

function NavBar() {
  return (
    <nav>
      <Link to="/">Home</Link>
    </nav>
  );
}

export default NavBar;
```

src/components/NavBar/NavBar.css

```
nav {
  background-color: #333;
  padding: 10px;
}

nav a {
  color: white;
  text-decoration: none;
  font-size: 18px;
}

nav a:hover {
  text-decoration: underline;
}
```

src/screens/Home/Home.jsx

```
import { useState } from 'react';
import { useDispatch, useSelector } from 'react-redux';
import { addBook } from '../../redux/slices/bookSlice';
import { addMember } from '../../redux/slices/memberSlice';
import BookList from '../../components/BookList/BookList';
import MemberList from '../../components/MemberList/MemberList';
import FilterButtons from '../../components/FilterButtons/FilterButtons';

function Home() {
  const dispatch = useDispatch();
  const [bookTitle, setBookTitle] = useState('');
  const [memberName, setMemberName] = useState('');

  const handleAddBook = () => {
    if (bookTitle.trim()) {
      dispatch(addBook({
        id: Date.now(),
        title: bookTitle,
        borrowed: false,
        borrowedBy: null,
        addedAt: new Date().toISOString(),
      }));
      setBookTitle('');
    }
  };
}
```

```

const handleAddMember = () => {
  if (memberName.trim()) {
    dispatch(addMember({
      id: Date.now(),
      name: memberName,
      joinedAt: new Date().toISOString(),
    }));
    setMemberName('');
  }
};

return (
  <div>
    <h1>Library Membership Manager</h1>
    <div>
      <h2>Add Book</h2>
      <input
        type="text"
        value={bookTitle}
        onChange={(e) => setBookTitle(e.target.value)}
        placeholder="Enter book title"
      />
      <button onClick={handleAddBook}>Add Book</button>
    </div>
    <div>
      <h2>Add Member</h2>
      <input
        type="text"
        value={memberName}
        onChange={(e) => setMemberName(e.target.value)}
        placeholder="Enter member name"
      />
      <button onClick={handleAddMember}>Add Member</button>
    </div>
    <FilterButtons />
    <BookList />
    <MemberList />
  </div>
);
}

export default Home;

```

src/components/BookList/BookList.jsx

```

import { useSelector } from 'react-redux';
import BookItem from '../BookItem/BookItem';
import './BookList.css';

function BookList() {
  const { books, filter } = useSelector(state => state.books);

  const filteredBooks = books.filter(book => {
    if (filter === 'all') return true;
    if (filter === 'borrowed') return book.borrowed;
    if (filter === 'available') return !book.borrowed;
    return true;
  });
}

```

```
return (  
  <div>  
    <h2>Books</h2>  
    <ul>  
      {filteredBooks.map(book => (  
        <BookItem key={book.id} book={book} />  
      ))}  
    </ul>  
  </div>  
}  
  
export default BookList;
```

src/components/BookList/BookList.css

```
ul {  
  list-style-type: none;  
  padding: 0;  
}
```

src/components/BookItem/BookItem.jsx

```
import { useDispatch, useSelector } from 'react-redux';  
import { Link } from 'react-router-dom';  
import { toggleBorrowed, removeBook } from '../../redux/slices/bookSlice';  
import './BookItem.css';  
  
function BookItem({ book }) {  
  const dispatch = useDispatch();  
  const members = useSelector(state => state.members.members);  
  
  const handleToggleBorrowed = (e) => {  
    const memberId = e.target.value === 'available' ? null : Number(e.target.value);  
    dispatch(toggleBorrowed({ bookId: book.id, memberId }));  
  };  
  
  return (  
    <li>  
      <Link to={`/${book.id}`}>{book.title}</Link>  
      <select value={book.borrowedBy || 'available'} onChange={handleToggleBorrowed}>  
        <option value="available">Available</option>  
        {members.map(member => (  
          <option key={member.id} value={member.id}>{member.name}</option>  
        ))}  
      </select>  
      <button onClick={() => dispatch(removeBook(book.id))}>Delete</button>  
    </li>  
  );  
}  
  
export default BookItem;
```

src/components/BookItem/BookItem.css

```
li {  
  display: flex;  
  align-items: center;
```

```
gap: 10px;
margin: 10px 0;
}

a {
  color: #007bff;
  text-decoration: none;
}

a:hover {
  text-decoration: underline;
}

button {
  background-color: #ff4444;
  color: white;
  border: none;
  padding: 5px 10px;
  cursor: pointer;
}

button:hover {
  background-color: #cc0000;
}
```

src/components/MemberList/MemberList.jsx

```
import { useSelector } from 'react-redux';
import MemberItem from '../MemberItem/MemberItem';
import './MemberList.css';

function MemberList() {
  const members = useSelector(state => state.members.members);

  return (
    <div>
      <h2>Members</h2>
      <ul>
        {members.map(member => (
          <MemberItem key={member.id} member={member} />
        ))}
      </ul>
    </div>
  );
}

export default MemberList;
```

src/components/MemberList/MemberList.css

```
ul {
  list-style-type: none;
  padding: 0;
}
```

src/components/MemberItem/MemberItem.jsx


```
import { useDispatch, useSelector } from 'react-redux';
import { Link } from 'react-router-dom';
import { removeMember } from '../redux/slices/memberSlice';
import './MemberItem.css';

function MemberItem({ member }) {
  const dispatch = useDispatch();
  const books = useSelector(state => state.books.books);
  const canDelete = !books.some(book => book.borrowedBy === member.id);

  const handleDelete = () => {
    if (canDelete) {
      dispatch(removeMember(member.id));
    } else {
      alert('Cannot delete member who is borrowing books.');
```

```
    }
  };

  return (
    <li>
      <Link to={`/${member.id}`}>{member.name}</Link>
      <button onClick={handleDelete} disabled={!canDelete}>Delete</button>
    </li>
  );
}
```

```
export default MemberItem;
```

src/components/MemberItem/MemberItem.css

```
li {
  display: flex;
  align-items: center;
  gap: 10px;
  margin: 10px 0;
}

a {
  color: #007bff;
  text-decoration: none;
}

a:hover {
  text-decoration: underline;
}

button {
  background-color: #ff4444;
  color: white;
  border: none;
  padding: 5px 10px;
  cursor: pointer;
}

button:disabled {
  background-color: #cccccc;
  cursor: not-allowed;
}
```

```
button:hover:not(:disabled) {
  background-color: #cc0000;
}
```

src/components/FilterButtons/FilterButtons.jsx

```
import { useDispatch } from 'react-redux';
import { setFilter } from '../../redux/slices/bookSlice';
import './FilterButtons.css';

function FilterButtons() {
  const dispatch = useDispatch();

  return (
    <div>
      <button onClick={() => dispatch(setFilter('all'))}>All</button>
      <button onClick={() => dispatch(setFilter('borrowed'))}>Borrowed</button>
      <button onClick={() => dispatch(setFilter('available'))}>Available</button>
    </div>
  );
}

export default FilterButtons
```

src/components/FilterButtons/FilterButtons.css

```
div {
  margin: 10px 0;
}

button {
  margin-right: 10px;
  padding: 5px 10px;
  background-color: #007bff;
  color: white;
  border: none;
  cursor: pointer;
}

button:hover {
  background-color: #0056b3;
}
```

src/screens/BookDetail/BookDetail.jsx

```
import { useParams, Link } from 'react-router-dom';
import { useSelector } from 'react-redux';

function BookDetail() {
  const { id } = useParams();
  const books = useSelector(state => state.books.books);
  const members = useSelector(state => state.members.members);
  const book = books.find(b => b.id === Number(id));

  if (!book) {
    return (
      <div>
```



```
    <h2>Book Not Found</h2>
    <Link to="/">Back to Home</Link>
  </div>
);
}

const borrower = book.borrowedBy ? members.find(m => m.id === book.borrowedBy) : null;

return (
  <div>
    <h1>Book Detail</h1>
    <p><strong>Title:</strong> {book.title}</p>
    <p><strong>Status:</strong> {book.borrowed ? 'Borrowed' : 'Available'}</p>
    {borrower && (
      <p><strong>Borrowed By:</strong> <Link to={` /member/${borrower.id}`}>{borrower.name}</Link></p>
    )}
    <p><strong>Added At:</strong> {new Date(book.addedAt).toLocaleString()}</p>
    <Link to="/">Back to Home</Link>
  </div>
);
}

export default BookDetail;
```

src/screens/MemberDetail/MemberDetail.jsx

```
import { useParams, Link } from 'react-router-dom';
import { useSelector } from 'react-redux';

function MemberDetail() {
  const { id } = useParams();
  const members = useSelector(state => state.members.members);
  const books = useSelector(state => state.books.books);
  const member = members.find(m => m.id === Number(id));

  if (!member) {
    return (
      <div>
        <h2>Member Not Found</h2>
        <Link to="/">Back to Home</Link>
      </div>
    );
  }

  const borrowedBooks = books.filter(book => book.borrowedBy === member.id);

  return (
    <div>
      <h1>Member Detail</h1>
      <p><strong>Name:</strong> {member.name}</p>
      <p><strong>Joined At:</strong> {new Date(member.joinedAt).toLocaleString()}</p>
      <h3>Borrowed Books</h3>
      <ul>
        {borrowedBooks.map(book => (
          <li key={book.id}>
            <Link to={` /book/${book.id}`}>{book.title}</Link>
          </li>
        ))}
      </ul>
    </div>
  );
}
```

```
    <Link to="/">Back to Home</Link>
  </div>
);
}

export default MemberDetail;
```

src/screens/NotFound/NotFound.jsx

```
import { Link } from 'react-router-dom';

function NotFound() {
  return (
    <div>
      <h1>404 - Page Not Found</h1>
      <Link to="/">Back to Home</Link>
    </div>
  );
}

export default NotFound;
```